

Q.166 | LC 2090: K Radius Subarray Averages

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Problem Intuition & Logic

For each index i , we need to calculate the average of:

- k elements on the left
- $\text{nums}[i]$ (the current element)
- k elements on the right

Total Window Size = $k + 1 + k = 2 * k + 1$

Approach

We use the **Standard Fixed Window Template** to solve this efficiently.

1. Initialize the result array `ans` of size n with -1 .
2. Calculate the `windowSize`. If it's strictly greater than the array size n , return `ans` immediately.
3. Compute the sum of the first window (from index 0 to `windowSize` - 1).
4. Assign the average of the first window to the center index (which initially is k).
5. Slide the window from $j = \text{windowSize}$ to $n - 1$:
 - Add the new element `nums[j]` to the sum.
 - Subtract the element that left the window `nums[j - windowSize]` from the sum.
 - Increment the center index.
 - Store the new window's average at the updated center.

C++ Implementation

```
class Solution {
public:
    vector<int> getAverages(vector<int>& nums, int k) {

        int n = nums.size();
        vector<int> ans(n, -1);

        int windowsize = 2 * k + 1;
        if(windowsize > n) {
            return ans;
        }
        long long sum = 0;
        for(int i = 0; i < windowsize; i++) {
            sum += nums[i];
        }
        int center = k;
        ans[center] = sum / windowsize;

        for(int j = windowsize; j < n; j++) {
            sum += nums[j];
            sum -= nums[j - windowsize];
            center++;
            ans[center] = sum / windowsize;
        }

        return ans;
    }
};
```